Hamilton Cycles in the Noncrossing Partition Refinement Graph

Alex Chengyu Li

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Abstract

Let $\mathrm{NC}_R(n)$ be the cover graph of the refinement order on the noncrossing partitions of an n-element cyclically ordered set. Thus an edge merges two blocks, or reverses such a merge by splitting one block. We determine exactly when this graph has a Hamilton cycle. For odd $n \geqslant 3$, block-count parity is a bipartition and a signed Dyck-tree recurrence gives color-class difference $(-1)^{m+1}\mathrm{Cat}_m$ at $n = 2m+1$, so a Hamilton cycle is impossible. For every even $n \geqslant 4$, we give one uniform construction. A noncrossing partition is encoded bijectively by a noncrossing partial matching Q and a Boolean mask on the unmatched nonroot points. Each fixed Q is therefore an odd-dimensional hypercube and carries a cyclic reflected Gray code. Deleting the lexicographically first chord makes the matchings into a rooted tree. For every tree edge we exhibit an explicit refinement diamond; an injective port assignment makes all selected cube edges disjoint. Recursive square switching then joins all Boolean cycles into one Hamilton cycle. With the formal convention for singleton graphs, the complete classification for all natural n is

$$\mathrm{NC}_R(n) \text{ is Hamiltonian} \iff n \in \{0,1\} \text{ or } (n \text{ is even and } n \geqslant 4).$$

The construction, the odd obstruction, the small cases, and the final classification are formalized in Lean 4. The publication theorem has no project axiom; its axiom audit is exactly `[propext, Classical.choice, Quot.sound]`.

Mathematics Subject Classifications: 05A18, 05C45, 05C85

1 Introduction

Write $X_n = \{0, 1, \ldots, n-1\}$ with its cyclic order. A partition of X_n is *noncrossing* if there are no $a < b < c < d$ for which a, c belong to one block and b, d belong to another. The set $\mathrm{NC}(n)$ of all noncrossing partitions is the Kreweras lattice under refinement [4, 8, 9]. Its cover graph $\mathrm{NC}_R(n)$ has vertex set $\mathrm{NC}(n)$; two vertices are adjacent when one is obtained from the other by merging exactly two blocks, with the result still noncrossing.

Independent Researcher, Tokyo, Japan (`cl7076@nyu.edu`).

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Why these are precisely the cover edges. The Kreweras lattice is graded by

$$\rho(\pi) = n - b(\pi),$$

where $b(\pi)$ is the number of blocks. If σ is obtained from π by merging two blocks, then $\pi < \sigma$ and $\rho(\sigma) = \rho(\pi)+1$, so no lattice element can lie strictly between them. Conversely, a cover raises the rank by one. Since every block of a coarsening is a union of blocks of the finer partition, lowering the number of blocks by one can only replace two blocks by their union. Thus the merge/split adjacency used below is exactly the cover relation, rather than a larger auxiliary graph.

The adjacency here is not the element-transfer adjacency in the cyclic Gray code of Huemer, Hurtado, Noy, and Omaña-Pulido [2]. The general permutation-language results of Hartung–Hoang–Mütze–Williams and Hoang–Mütze [1, 3] provide Hamilton paths for many quotient graphs, but do not identify the refinement cover graph with such a quotient. The arbitrary-sublist refinement Gray-code problem considered by Merino, Namrata, and Williams [5] is also different from the spanning cycle problem for the full lattice. For broader background on combinatorial Gray codes and their Hamiltonian formulations, see Mütze's survey [6].

Petersen [7] constructed a symmetric Boolean decomposition of the noncrossing partition lattice using valley hopping. Our proof makes the corresponding coordinates direct: a Boolean block is indexed by a canonical noncrossing partial matching, and its Boolean coordinate is an explicit mask. This direct description is the one used in the machine proof.

Our main result is the following complete classification.

Theorem 1 (Complete classification)**.** *Under the closed-spanning-walk convention used in the formalization, for every $n \in \mathbb{N}$,*

$$\mathrm{NC}_R(n) \text{ has a Hamilton cycle} \iff n = 0 \text{ or } n = 1 \text{ or } (n \text{ is even and } 4 \leqslant n).$$

Consequently, in the usual nondegenerate range $n \geqslant 3$, $\mathrm{NC}_R(n)$ has a Hamilton cycle if and only if n is even.

Sections 2 and 3–5 prove the two directions. Section 7 gives the exact correspondence between the displayed results and Lean declarations.

2 The odd-order obstruction

Let $b(\pi)$ denote the number of blocks of π. Every cover changes $b(\pi)$ by one, hence block-count parity colors the graph properly.

Lemma 2. *The graph $\mathrm{NC}_R(n)$ is bipartite, with color classes*

$$E_n = \{\pi : b(\pi) \text{ is even}\}, \qquad O_n = \{\pi : b(\pi) \text{ is odd}\}.$$

Proof. A cover relation merges two blocks or splits one block into two. Thus the two endpoints have block counts differing by exactly one, and therefore have opposite parity. $\square$

Define the signed count

$$D_n := \sum_{\pi \in \mathrm{NC}(n)} (-1)^{b(\pi)} = |E_n| - |O_n|.$$

Under the standard Dyck-word, or equivalently plane-binary-tree, correspondence for noncrossing partitions, the number of blocks is the number of peaks. Decomposing a nonempty tree at its root yields the recurrence

$$D_{n+1} = -D_n + \sum_{k=0}^{n-1} D_{k+1} D_{n-1-k}, \qquad D_0 = 1, quad D_1 = -1. \tag{1}$$

Theorem 3 (Signed count)**.** *For $m \geqslant 1$,*

$$D_{2m} = 0, \qquad D_{2m+1} = (-1)^{m+1} \mathrm{Cat}_m,$$

where $\mathrm{Cat}_m = \frac{1}{m+1}\binom{2m}{m}$.

Proof. We use simultaneous strong induction on m. In (1) for D_{2m}, the two indices in every product sum to $2m-1$. Hence one index is positive and even, except for the boundary term $D_{2m-1}D_0$. Every nonboundary term vanishes by induction, and the boundary term cancels the initial $-D_{2m-1}$. Thus $D_{2m} = 0$.

For D_{2m+1}, all terms having a positive even index vanish. The remaining terms have indices $2j+1$ and $2(m-1-j)+1$. By induction their product is

$$(-1)^{j+1} \mathrm{Cat}_j (-1)^{m-j} \mathrm{Cat}_{m-1-j} = (-1)^{m+1} \mathrm{Cat}_j \mathrm{Cat}_{m-1-j}.$$

The initial term is zero because $D_{2m} = 0$. Summing and using the Catalan recurrence $\mathrm{Cat}_m = \sum_{j=0}^{m-1} \mathrm{Cat}_j \mathrm{Cat}_{m-1-j}$ proves the formula. The initial values $D_0 = 1$ and $D_1 = -1$ start the induction. $\square$

Theorem 4 (Odd-order obstruction)**.** *If $n \geqslant 3$ is odd, then $\mathrm{NC}_R(n)$ has no Hamilton cycle.*

Proof. Write $n = 2m+1$. Since $n \geqslant 3$, $m \geqslant 1$, and Theorem 3 gives $|E_n| - |O_n| = (-1)^{m+1}\mathrm{Cat}_m \neq 0$. Every cycle in a bipartite graph uses equally many vertices from its two color classes. A spanning cycle would therefore force $|E_n| = |O_n|$, a contradiction. $\square$

3 Canonical matching–mask coordinates

We now assume $n > 0$. A *canonical matching* is a set Q of ordered chords (a, b) with

$$0 < a < b < n,$$

such that distinct chords have disjoint endpoints and no two chords cross. The chords may be nested. Let $\mathrm{end}(Q)$ be their endpoint set and define the free points

$$F(Q) = X_n \setminus (\{0\} \cup \mathrm{end}(Q)).$$

For a chord endpoint x, let $e_Q(x)$ be the left endpoint of its chord. For $x \in F(Q)$, let $a_Q(x)$ be the left endpoint of the innermost chord whose open interval contains x, or 0 if there is no such chord.

For a mask $A \subseteq F(Q)$ define

$$\kappa_{Q,A}(x) = \begin{cases} e_Q(x), & x \in \mathrm{end}(Q), \\ a_Q(x), & x \in A, \\ x, & \text{otherwise.} \end{cases} \tag{2}$$

Let $P_Q(A)$ be the partition into the fibers of $\kappa_{Q,A}$.

Lemma 5 (Key-fiber geometry)**.** *If two distinct points have the same decoded key, then that key is either* 0 *or the left endpoint of a chord of Q. A fiber based at a chord (a, b) is contained in $[a, b]$. A point in that fiber cannot lie strictly inside a properly nested chord with a different left endpoint, and a point in the root fiber cannot lie strictly inside any chord of Q. Consequently, two distinct decoded fibers cannot cross.*

Proof. The first assertion follows directly from (2). A point can change its own key only by being a chord endpoint or by being selected in the mask. In the first case the common key is that chord's left endpoint. In the second it is the left endpoint of the innermost containing chord, or 0.

Fix $(a, b) \in Q$. Its endpoints decode to a. If another point x also decodes to a, then either x is one of these endpoints or x is selected and (a, b) is its innermost containing chord; in either case $a \leqslant x \leqslant b$. If (c, d) is properly nested in (a, b) and $c < x < d$, then (c, d) is a strictly inner containing chord, so the anchor of a selected x is at least $c > a$; hence x cannot decode to a. The same argument with no outer chord shows that a root-fiber point cannot lie inside any chord.

Suppose fibers based at r and s crossed, with $x_1 < y_1 < x_2 < y_2$ in the two fibers. If one key is 0, the other fiber's chord has both endpoints between two root-fiber points, contradicting the root assertion. Otherwise the two base chords are disjoint or nested because Q is noncrossing. The four alternating points force one fiber point into a properly nested chord with a different base, contradicting the preceding paragraph. ☐

Theorem 6 (Boolean decomposition)**.** *For every $n > 0$, the map*

$$(Q, A) \longmapsto P_Q(A), \qquad Q \text{ canonical}, \quad A \subseteq F(Q),$$

is a bijection onto $\mathrm{NC}(n)$. *Moreover, if $q \in F(Q) \setminus A$, then $P_Q(A \cup \{q\})$ covers $P_Q(A)$ in the refinement lattice.*

Proof. First note that $\kappa_{Q,A}(x) \leqslant x$. By Lemma 5, four alternating points cannot lie in two different fibers. Thus $P_Q(A)$ is noncrossing.

Conversely, for $\pi \in \mathrm{NC}(n)$, take from every nonsingleton block not based at 0 the chord joining its minimum and maximum. Distinct blocks give endpoint-disjoint noncrossing chords, so this is a canonical matching $Q(\pi)$. Put in $A(\pi)$ exactly those remaining nonroot, nonendpoint points whose block is nontrivial. The minimum of the block containing such a point is precisely the left endpoint of the innermost containing chord, or 0. Therefore

$$\kappa_{Q(\pi),A(\pi)}(x) = \min(\text{the block of } x)$$

for every x, and hence $P_{Q(\pi)}(A(\pi)) = \pi$.

The same description recovers Q from the extrema of the nonroot nonsingleton fibers of $P_Q(A)$, and then recovers A by testing which free points have nonsingleton fibers. The decoder is therefore injective and surjective. Finally, inserting q into the mask changes only its key, from q to $a_Q(q)$; it merges the singleton block $\{q\}$ with the block based at $a_Q(q)$ and leaves every other block unchanged. This is one refinement cover. $\square$

The dimension of the block indexed by Q is

$$d(Q) := |F(Q)| = n - 1 - 2|Q|. \tag{3}$$

If n is even, then $d(Q)$ is positive and odd.

4 Gray cycles, the block tree, and ports

Order $F(Q)$ increasingly and use the binary-reflected Gray list on its subsets. Adjacent masks differ by one free point, so Theorem 6 sends every Gray edge to a refinement edge.

Lemma 7 (Block Gray cycle)**.** *For even n, every block $\{P_Q(A) : A \subseteq F(Q)\}$ has a cyclic Gray listing that visits each of its vertices once. For $d(Q) \geqslant 3$ it is a simple cycle; for $d(Q) = 1$ it is the two directed occurrences of the unique edge, to be consumed by the tree switch below.*

Proof. The reflected recursion

$$G(x_0, x_1, \ldots) = G(x_1, \ldots), \mathrm{rev}(G(x_1, \ldots)) + x_0$$

lists every subset once, changes one coordinate at every step, and has a closing one-coordinate change. Equation (3) makes the dimension positive. Decoding gives the asserted refinement edges and the bijection gives distinctness and coverage. $\square$

If $Q \neq \varnothing$, let $p(Q)$ be obtained by deleting the lexicographically first chord of Q.

Lemma 8 (Canonical block tree)**.** *The map p defines a rooted tree on canonical matchings, with root the empty matching. Each parent step decreases chord count by one and increases free dimension by two. Every canonical matching lies in the root subtree.*

Proof. Deleting a chord preserves endpoint disjointness and noncrossingness. It decreases $|Q|$ by one, so iteration terminates at the unique chordless matching. The parent is uniquely determined by the lexicographically first chord. Conversely, every nonroot matching has exactly that parent. Thus the directed parent relation is acyclic, connected to the empty matching, and has unique parent at every nonroot vertex; it is a rooted tree. Formula (3) gives the dimension change. □

Let C be a child of Q, obtained by inserting its first chord between the ith and jth free points of Q, $i < j$. The reflected Gray cycles admit the following explicit opposite edges. Indices in a mask refer to the increasing free-point order of the corresponding block.

(i,j)	parent edge in Q	child edge in C
$1 \leqslant i < j$	$\{i,j\}, \{0,i,j\}$	$\varnothing, \{0\}$
$i = 0,\ j \geqslant 2$	$\{0,j\}, \{0,1,j\}$	$\varnothing, \{0\}$
$(i,j) = (0,1)$	$\varnothing, \{d(Q)-1\}$	$\varnothing, \{d(C)-1\}$.

(4)

Lemma 9 (Explicit joining diamond)**.** *For every parent–child pair Q, C, the two displayed Gray edges in the appropriate row of* (4)*, together with two cross-block refinement edges, form a four-cycle in* $\mathrm{NC}_R(n)$*.*

Proof. Write the free points of the parent as $u_0 < u_1 < \cdots < u_{d(Q)-1}$, and let the deleted chord be (u_i, u_j). The free points of the child are the remaining u_r's, in the same order. If B is a child mask, let $\widehat{B}$ be its order-preserving lift to the parent coordinates. Except in the special third row, the parent mask paired with B is

$$M(B) = \widehat{B} \cup \{i,j\}. \tag{5}$$

The decoder gives a direct comparison. In the child, u_i and u_j are the endpoints of one matching chord and hence have the same decoded key. In the parent they are free, and selecting both in $M(B)$ sends them to their common containing anchor. All other selected points have the same anchor before and after the order-preserving lift, while every unselected free point keeps its own key. More precisely, if r is the anchor of u_i after the chord is deleted, then

$$\kappa_{Q,M(B)}(x) = \begin{cases} r, & \kappa_{C,B}(x) = u_i, \\ \kappa_{C,B}(x), & \kappa_{C,B}(x) \neq u_i. \end{cases}$$

Here $r < u_i$, so the child fibers with keys r and u_i are distinct. Thus $P_Q(M(B))$ is obtained from $P_C(B)$ by replacing precisely those two fibers by their union, and the pair is joined by one cover edge.

We now verify that the four masks in each row of (4) are the ones just described and that the horizontal pairs are Gray edges.

If $i > 0$, the least child free point lifts to u_0. Hence

$$M(\varnothing) = \{i, j\}, \qquad M(\{0\}) = \{0, i, j\}.$$

The parent pair flips coordinate 0, and the child pair $\varnothing, \{0\}$ also flips coordinate 0. The cross covers are

$$P_C(\varnothing) \lessdot P_Q(\{i, j\}), \qquad P_C(\{0\}) \lessdot P_Q(\{0, i, j\});$$

these are the first row.

If $i = 0$ and $j \geqslant 2$, the least child free point lifts to u_1. Thus

$$M(\varnothing) = \{0, j\}, \qquad M(\{0\}) = \{0, 1, j\}.$$

The parent pair flips coordinate 1, while the child pair again flips its coordinate 0. The cross covers are

$$P_C(\varnothing) \lessdot P_Q(\{0, j\}), \qquad P_C(\{0\}) \lessdot P_Q(\{0, 1, j\});$$

these are the second row.

Finally suppose $(i, j) = (0, 1)$. Here the two selected horizontal edges are the closing Gray edges. The rightmost child free point lifts to the rightmost parent free point. Directly from (2), inserting the chord merges exactly the two endpoint fibers both at the empty mask and at the rightmost singleton mask. Hence the cross covers are

$$P_Q(\varnothing) \lessdot P_C(\varnothing), \qquad P_Q(\{d(Q) - 1\}) \lessdot P_C(\{d(C) - 1\}).$$

The two horizontal pairs are the closing edges in their respective block cycles. In all three cases the four vertices are distinct across the two Boolean blocks, both horizontal pairs are fixed block-cycle edges, and both vertical pairs are single merges. They therefore form the required literal four-cycle. □

For bookkeeping, call a Gray edge's *port key* either (q, H), meaning "flip coordinate q above ambient lower mask H", or *closing*. The three rows of (4) give one incoming key in the parent and one outgoing key in the child.

Lemma 10 (Port injection)**.** *For a fixed block Q, distinct children are assigned to distinct directed edge occurrences of the Gray cycle of Q. If Q is nonroot, its outgoing edge occurrence toward $p(Q)$ is not assigned to any child.*

Proof. Record a nonclosing directed Gray occurrence by a pair (q, H), where q is the flipped ambient point and H is the lower ambient mask. Equality of two such keys forces equality of both q and H; the closing occurrence is a separate key.

Let C be a child of Q, with first chord $e = (u_i, u_j)$. The parent-side key supplied by (4) is

$$\begin{cases} (u_0, \{u_i, u_j\}), & i > 0, \\ (u_1, \{u_0, u_j\}), & i = 0,\ j \geqslant 2, \\ \text{closing}, & (i, j) = (0, 1). \end{cases}$$

In each of the first two cases the lower mask is exactly the two-element endpoint set of e. Because the endpoints are ordered increasingly, that set uniquely recovers the ordered chord. The child itself is then recovered as $Q \cup \{e\}$. Thus two equal coordinate keys come from the same child. If both keys are closing, both first chords join the first two free points of Q, so again the chords and children coincide. A coordinate key cannot equal the closing key. This proves injectivity.

It remains to check that the edge retained for the parent of Q is not used by a child of Q. If the outgoing key is a coordinate key, its lower mask is $\varnothing$, whereas every incoming coordinate key has lower mask $\{u_i, u_j\} \neq \varnothing$; an incoming closing key has a different key type. If the outgoing key is closing, the first chord of Q joins its first two parent-free points. Any child chord would have both endpoints to the right of that chord, but would also have to be lexicographically no later than it in order to become the child's first chord, a contradiction. Hence in this special case Q has no children at all. The outgoing occurrence is therefore unassigned in every case. □

5 Recursive square switching

We use the elementary square switch: if two disjoint cycles contain opposite edges ab and cd of a four-cycle $abdca$, deleting ab, cd and adding ac, bd produces one cycle on the union of their vertices. Applying switches sequentially is delicate because a parent cycle is modified by its earlier children. The following recursive formulation records exactly which edge occurrence remains available.

Lemma 11 (Block-tree assembly)**.** *For every canonical matching Q and even n, there is a cycle whose support is exactly the union of all Boolean blocks in the subtree rooted at Q. If Q is nonroot, the cycle retains the outgoing port prescribed by the diamond joining Q to $p(Q)$.*

Proof. We prove simultaneously the following three assertions for the subtree at Q:

1. there is one simple cycle Z_Q;

2. its support is exactly the union of the Boolean blocks in the subtree;

3. if Q is nonroot, the prescribed outgoing directed edge of the block cycle of Q is still an edge of Z_Q.

Induct on $d(Q) = |F(Q)|$. A child is obtained by adding one chord and hence has dimension $d(Q) - 2$, so this is a well-founded induction. Assume the three assertions for every child.

Write the cyclic Gray list of the block of Q as $v_0, v_1, \ldots, v_{r-1}$, with indices modulo r, and regard $e_k = (v_k, v_{k+1})$ as a directed edge occurrence. By Lemma 10, each child is assigned to a distinct e_k. For an unassigned occurrence e_k, define the corresponding piece to be the singleton list $[v_k]$. For an occurrence assigned to a child C, the inductive cycle Z_C still contains its prescribed outgoing edge ab. The joining diamond has opposite edges

e_k and ab. Delete those two edges and use the two cross edges of the diamond. Cutting Z_C at ab turns it into a path through every vertex of the child subtree; adjoining v_k and the two cross edges gives a path piece whose entry is v_k and whose exit is adjacent to v_{k+1}. Thus every piece associated with e_k starts at the source of e_k and hands off to the source of e_{k+1}.

We verify that concatenation introduces neither omissions nor repetitions. The vertices v_k are distinct because the block Gray list is a cyclic listing. A child-subtree piece contains no vertex in the block of Q: the canonical matching of every one of its vertices lies strictly below that child in the matching tree, whereas the vertices v_k have canonical matching exactly Q. Two different child-subtree pieces are disjoint because a rooted tree has disjoint subtrees below distinct children. Finally, port injectivity ensures that no child subtree is inserted twice. Therefore all pieces are nonempty, internally simple, and pairwise disjoint.

Their internal edges are either edges of a cut child cycle or one of the two cross edges in its joining diamond. The exit of the kth piece is adjacent to v_{k+1}, the head of the next piece; the final piece hands off to v_0 in exactly the same way. Flattening the cyclic sequence of pieces is thus a single simple cycle.

Every vertex in the block of Q occurs as exactly one v_k. Every proper descendant belongs to the subtree of a unique child, and that entire child subtree occurs in the unique piece assigned to that child. The support is therefore exactly the subtree of Q. If Q is nonroot, its outgoing occurrence is unassigned by Lemma 10; its piece is the singleton $[v_k]$, so the handoff edge $v_k v_{k+1}$ remains present in the flattened cycle. This proves the third assertion and closes the induction. □

Theorem 12 (Uniform even-order construction)**.** *For every even $n \geqslant 4$, $\mathrm{NC}_R(n)$ has a Hamilton cycle.*

Proof. Apply Lemma 11 to the empty matching. By Lemma 8, every canonical matching lies in its subtree. By Theorem 6, the corresponding Boolean blocks partition all of $\mathrm{NC}(n)$. The assembled root cycle is therefore spanning. Since $n \geqslant 4$, $|\mathrm{NC}(n)| = \mathrm{Cat}_n \geqslant 3$, so the closed spanning walk is a Hamilton cycle in the simple graph. □

The proof is uniform at $n = 4$ and $n = 6$; no exhaustive computation is used for either order.

6 Proof of the classification

Proof of Theorem 1. For even $n \geqslant 4$, apply Theorem 12. For odd $n \geqslant 3$, apply Theorem 4.

It remains to inspect $n < 4$. There is one noncrossing partition for $n = 0$ and one for $n = 1$; the formal closed-spanning-walk convention regards each singleton graph as Hamiltonian. For $n = 2$, the refinement graph consists of one edge on two vertices and has no simple Hamilton cycle. For $n = 3$, the odd obstruction applies (equivalently, the five-vertex bipartition is unbalanced). These cases give exactly the stated disjunction. □

7 Machine-checked correspondence

The accompanying Lean 4 package follows the proof above rather than the chronology of its discovery. Table 1 lists the exact declarations consumed by each paper statement. The file `paper/theorem-map.json` is the machine-readable source of this table, and the package checker verifies that every paper label and every declaration is present.

Table 1: Paper statements and Lean declarations.

Paper label	Lean declarations
`lem:bipartite`	`NCRefinementGraph_isBipartite`
`thm:signed-count`	`signedDyckSumTree_closed_form`, `signedNCCount_univ_eq_signedDyckSumTree`
`thm:odd-obstruction`	`NCRefinementGraph_fin_odd_geq3_not_isHamiltonian`
`lem:key-fibers`	`commonKey_root_or_chord`, `point_in_chord_hull`, `point_avoids_nested_chord`, `root_point_avoids_chord`
`thm:boolean-decomposition`	`decodeState_bijective`, `stateEquivNC`, `decodedNC_mergesTo_insert`
`lem:block-gray-cycle`	`rankGrayCycle`, `rankGrayVertices_nodup`, `rankGrayVertices_chain`
`lem:canonical-block-tree`	`parent_card`, `parentOption_eq_none_iff`, `inSubtree_root`
`lem:joining-diamond`	`joiningDiamond_nonempty`, `joiningDiamond`
`lem:port-injection`	`portAssignedToPair_exists`, `assignedPair_injective`, `outgoingPair_not_assigned`
`lem:tree-assembly`	`EdgePieceSystem.toCycleCode`, `subtreeCycle`, `rootCycleCode_support`
`thm:uniform-even`	`NCRefinementGraph_fin_even_geq4_isHamiltonian_petersen`
`thm:classification`	`NCRefinementGraph_fin_isHamiltonian_iff`

The publication root is `Hamilton.HamiltonianCycles`. The command `lake env lean Hamilton/HamiltonianCyclesAxiomAudit.lean` reports for the uniform even theorem, the odd obstruction, and the complete classification exactly

```
[propext, Classical.choice, Quot.sound].
```

These are standard Lean/Mathlib logical dependencies. No project axiom, `sorry`, `admit`, or `native_decide` occurs in the publication dependency closure.

The kernel-checked formalization and the initial technical manuscript snapshot were archived as version v1.0.0 at doi:10.5281/zenodo.21316751; the version-independent DOI doi:10.5281/zenodo.21316750 resolves to the latest archive version. The present manuscript revision adds expository proof detail without changing any formal statement or Lean source.

8 Concluding remarks

The obstruction and construction meet exactly. Odd orders fail for the global reason forced by the bipartition sizes. Even orders admit a cycle because every Petersen block has odd Boolean dimension and because the lexicographic chord tree admits conflict-free explicit joining ports. The central point is that the proof never asks for a Hamilton path with arbitrary prescribed endpoints inside an opaque fiber. Instead, its endpoints are fixed Gray edges, and the cross-block edges are constructed together with those ports in a literal refinement diamond.

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